THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 6: **Heat Transfer in Nature** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Two rods A and B of equal size are heated at one end. A's pins fall in 20 s, B's pins fall in 60 s. This shows that: 1

(a) A is a better conductor **(b)** B is a better conductor
(c) both equal **(d)** neither conducts

2. At a seaside at noon, the sea breeze blows. At 11 p.m. the breeze direction is: 1

(a) reversed **(b)** upward **(c)** zero always **(d)** same

3. Which of the following shows a correct application of convection? 1

(a) spoon heating **(b)** cloth drying **(c)** sunlight on skin **(d)** sea breeze

4. A student places a black can and a white can with equal water in the sun. After 30 min the black can's water is hotter because: 1

(a) it has more water **(b)** it reflects more
(c) it conducts less **(d)** it absorbs more radiation

5. At the junction between the two breezes at around dusk, the wind: 1

(a) reverses at once strongly **(b)** doubles
(c) becomes a storm **(d)** nearly stops

6. Which of the following is a correct application of conduction? 1

(a) sea breeze **(b)** heating iron rod **(c)** ceiling fan **(d)** sunlight

7. A candle is placed under a paper spiral hung from a thread. The spiral rotates because: 1

(a) hot air rising pushes it **(b)** heat radiation pushes
(c) cold air pushes it **(d)** wind from the room

8. If the sea and the land were heated equally, the breezes would: 1

(a) be stronger **(b)** be of storms **(c)** not form **(d)** be reversed

9. **Assertion (A):** Heat always flows from a hotter body to a colder body. 1
Reason (R): Heat flows until the two bodies reach the same temperature.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** The heat of the Sun reaches the Earth by radiation. 1
Reason (R): Conduction and convection need a material medium, which is almost absent between the Sun and the Earth.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Explain why heating a liquid from the top does not heat it quickly but from the bottom does. 2

12. Why do we place a cloth wet with water around a matka in summer? 2

13. Why does ice wrapped in a blanket not melt for long? 2

14. Why are the fins of a radiator painted black and made of metal? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In an experiment, 100 mL of hot water at 80 °C is poured into three cups: metal, ceramic and thermocol. After 10 min the temperatures are 55 °C, 62 °C and 72 °C respectively. Arrange them and explain. 3

16. Explain why two spoons, one of steel and one of plastic, in a cup of boiling water show different results. 3

17. Explain why woollen blankets keep you warm but do not generate heat. 3

OR

Explain how the design of a house with high ceilings, small windows and thick walls helps in hot regions. 3

18. Explain how a refrigerator removes heat from inside to outside (in simple words). 3

OR

A metal and a wooden spoon at 25 °C are placed in boiling water. Predict the temperature of the handles after 1 min. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Sunita is cooking dal on a gas stove. She leaves a steel spoon in the pan and, after some time, the spoon's handle becomes too hot to hold. She uses a wooden spoon to stir. She also sees the dal circulating inside the pan as it boils. 5
(i) By which mode does the handle of the steel spoon get hot?
(ii) Why is the wooden spoon preferred?
(iii) By which mode does the dal circulate?
(iv) Explain how the water in a pan gets heated from the bottom to the top.

20. Explain how a solar cooker works and why each part is designed as it is. 5

OR

Discuss how knowledge of heat transfer is used in designing winter clothing, summer clothing and food storage. 5

END OF QUESTION PAPER — Total: 40 Marks